Supplementary File S5: Experimental Protocol and Evidence Ledger

This supplement documents the latest available system-integration evidence and its limits. The runs below were performed against local virtual industrial endpoints. Evidence units are kept separate because they differ in task, actor, runtime configuration, and endpoint state. The associated machine-readable records and original CSV exports are supplied in `Supplementary-File-S5-Evidence.zip`. A companion HTML audit report provides an interactive step-by-step view.

I. Evidence Scope and Separation

The primary case is the completed 2026-10-31 document-driven HTTP/MQTT integration. It evaluates interface mapping, acquisition, read-only lead-task execution, and export consistency. The setpoint write/readback check was operator initiated and is reported separately from the agent task. Separate protocol and process-mission records are independent supporting evidence; counts and trajectories are not pooled. No physical plant was connected.

II. Document-Driven Virtual Integration

A. Configuration and execution

An interface contract declared six signals: HTTP enclosure temperature PV, setpoint, setpoint readback, and a 64-point thickness profile; and MQTT ambient temperature and humidity. The configuration created one new line, five DAQ nodes, one DCW node, a product, and a recipe. Provisioning reported 11/11 checks passed and zero cross-line write-point conflicts. The resulting Channel contained a lead and three configured worker roles. The archived activation evidence verifies only the lead as wired, and the retained task was assigned to that lead. The task was read-only and terminated `COMPLETED` at 100%.

B. Acquisition and export

The lead task queried five acquisition streams: four scalar signals and the vector profile. The task record contains 17 aggregated buckets for each scalar signal and 22 profile frames, each with 64 points. The observed ranges and export checks are summarized in Table I. Scalar query buckets and exported CSV rows have different aggregation semantics and must not be added together. The 19/19 export checks verify headers, manifest counts, value ranges, and vector shape. The CSV archive includes startup rows marked `offline`; thus, contract consistency does not imply that every row is a valid online observation.

TABLE I
Read-only task observations and exported data.

Signal	Task coverage	Observed range or structure
Enclosure PV	17 buckets	24.515–25.480 °C; mean 25.052 °C
SP readback	17 buckets	25.0 °C throughout the task window
Thickness profile	22 frames	64 points/frame; frame means 2.4975–2.5041 mm
Ambient temperature	17 buckets	20.5–23.44 °C
Ambient humidity	17 buckets	50.0–59.8%
Scalar CSVs	818 rows	Four files; includes startup offline rows
Vector CSV	3,456 rows	54 frames × 64 points/frame
Export validation	19/19	Manifest, headers, counts, ranges, and shape checks

C. Independent write/readback check

The virtual line began at a recipe setpoint of 25 °C. An early operator request for 27 °C was rejected because the 60-s write interval had not elapsed. A later operator/controller check wrote 27 and restored 25. A separate DAQ node observed both values. The HTTP write response echoed the request and is not treated as device readback. The write records have no task, approval, actor, or driver-event identifier linking them to the read-only agent task. This sequence therefore demonstrates an independent simulator write/readback path, not agent-controlled actuation.

III. Separate Protocol-Matrix Record

An earlier deterministic protocol matrix covered five local simulator protocols, 52 acquisition samples, and four write-capable protocol paths; all 24 out-of-window probes were rejected. The LLM loop was disabled. This run is reported as a separate protocol/guard check and is not pooled with the document-driven case or the latest integrated benchmark.

IV. Four Scripted Process Missions

The September 28 mission records provide additional multi-scene simulator coverage. Each row is one deterministic trajectory; scripted routines, rather than an LLM, selected setpoints. These trajectories are not replications of the October document-driven integration and do not isolate causal action effects. Some terminal secondary variables were not archived.

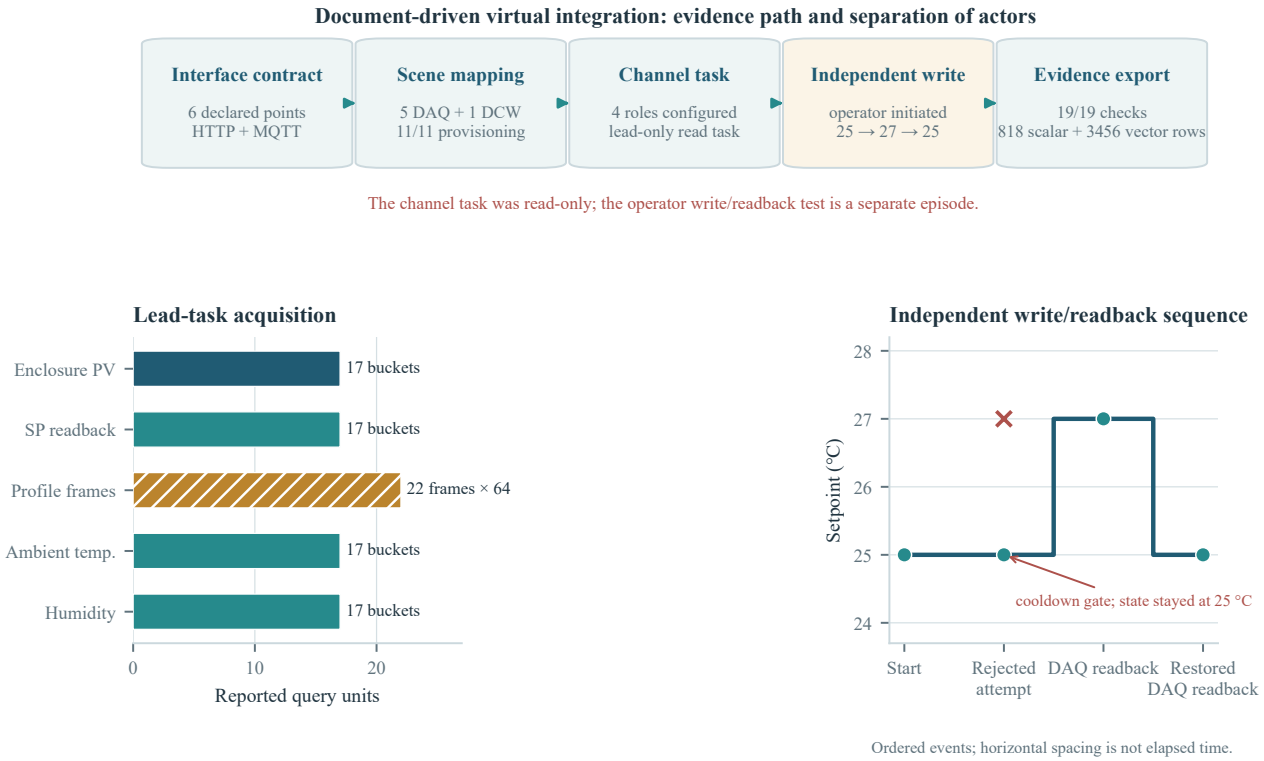


Fig. 1. Evidence sequence for the document-driven virtual-line case. The read-only lead task and operator-initiated writes are intentionally shown as separate paths. The lower panels summarize the retained scalar buckets, vector frames, and ordered setpoint observations. Horizontal spacing denotes event order only, not elapsed time or a causal trajectory.

TABLE II
REPORTED PRIMARY VALUES IN THE FOUR SCRIPTED MISSIONS.

Scene	Target band; initial → reported final
Injection	32.15–32.85 g; 31.296 → 32.36 g
BOPET film	24.3–25.7 μm; 28.03 → 25.30 μm
Annealing	89–101 HV; 134.9 → 99.5 HV
Wastewater	2.8–4.0 mg/L DO; 0.194 → 3.54 mg/L

V. Evidence Completeness and Manuscript Use

The primary integration case supports a bounded claim that a declared HTTP/MQTT virtual scene can be mapped into configured acquisition/control nodes, queried by a completed lead task, and exported with checked manifest/vector structure. The separate DAQ observations support one operator-initiated simulator write/readback demonstration. These data do not demonstrate multi-agent collaboration, approval-only causal effects, harness/model equivalence, repeated-run process performance, or physical deployment.

The evidence ZIP contains selected task, provisioning, readback, export, and process-mission records together with the raw CSV exports. The local temporary path in the export-statistics record is redacted; its original and delivered SHA-256 hashes are listed in `S5-EVIDENCE-MANIFEST.json`, and the delivered-file hashes are repeated in `SHA256SUMS`. The package hashes establish integrity of the delivered copy, not a run-time-signed provenance chain.